

Assignment, Wages, Platforms, And Pricing Rules

Algorithmic Rules As Labor-Market Institutions

Special Topic 7: Labor Market Design, Contracting, and Mechanism Design – Week 4

Central question

- How do assignment algorithms, platform rules, and wage-setting systems shape labor allocation and worker welfare?
- Week 4 moves from contracts inside firms to rules chosen by labor-market intermediaries.
- The labor objects are visibility, assignment, wages, commissions, monitoring, reputation, flexibility, risk, and future access to work.
- The research standard is rule $\rightarrow$ labor margin $\rightarrow$ counterfactual design $\rightarrow$ welfare incidence.

Intermediary objective and rule framework

$$\rho = (V, A, P, M)$$

- V : visibility, search order, ranking, recommendations, invitations.
- A : matching, routing, assignment, scheduling, queue discipline.
- P : posted wage, bids, commissions, floors, bonuses, dynamic prices, transparency.
- M : monitoring, ratings, guarantees, dispute rules, deactivation, evaluation.

$$\max_{\rho} \Pi(\rho) = \text{fees} + \text{future thickness} + \text{trust} - \text{operational and legal costs}$$

- The platform's objective can diverge from worker surplus or social surplus.

Worker welfare under platform rules

$$W_i(\rho) = \mathbb{E}[w_i(\rho)] - c_i(e_i, a_i) - \lambda_i \text{Var}(w_i(\rho)) + \phi_i \text{Flex}_i(\rho) + q_i(\rho) - s_i(\rho)$$

- Earnings are only one part of platform worker welfare.
- Rules affect time costs, volatility, flexibility, match quality, search burden, and monitoring pressure.
- A design can raise fill rates and still shift risk or bargaining power onto workers.
- Welfare claims need the relevant worker margin, not just platform output.

Assignment, visibility, and ranking

Rule	Observable margin	Interpretation risk
Search ranking	applications, invitations, hires	attention versus quality
Recommendation	employer contact, match rate	platform steering versus fit
Public signal	hire rate, wages, future jobs	information versus exposure
Guarantee	buyer trust, selected workers	credibility versus favoritism

- Workers and employers rarely see a neutral market menu.
- Good papers separate mechanical visibility, belief updating, and equilibrium response.

Wages, prices, commissions, and transparency

- Compensation design includes posted wages, bargaining, bids, commissions, wage floors, surge pricing, bonuses, and pay transparency.
- Wage information changes search targeting, applicant composition, employer screening, and accepted pay.
- Commissions and dynamic prices can allocate workers across times and tasks, but also change risk incidence.
- Pay-rule experiments are powerful because the rule itself is varied, not inferred from equilibrium wages alone.
- The key question is who absorbs the incidence: workers, employers, consumers, or the platform.

Worker welfare, flexibility, and risk

- Flexibility can be a worker amenity when workers control timing and task choice.
- Flexibility can also be a risk-shifting channel when workers absorb waiting time, cancellations, low demand, or algorithmic schedule pressure.
- Platform welfare objects include earnings, volatility, unpaid time, autonomy, ratings, deactivation risk, and future access to work.
- Average hourly pay is incomplete if task availability, risk, and continuation value change at the same time.
- Ride-hailing and online labor markets make these nonwage margins empirically visible.

Theory-to-applied translation

- ① Rule: visibility, assignment, disclosure, wage floor, commission, guarantee, monitoring, or deactivation.
- ② Labor margin: applications, invitations, hires, wage bids, accepted wages, hours, ratings, retention, or future jobs.
- ③ Counterfactual: what would users have seen, paid, earned, or faced under the old rule?
- ④ Mechanism: information, attention, sorting, bargaining power, risk allocation, or platform steering.
- ⑤ Welfare object: worker surplus, match quality, employer surplus, platform revenue, fairness, flexibility, or access.

Frontier experimental methods

Design	Best use
Platform A/B tests	ranking, visibility, recommendations, assignment
Information disclosure	ratings, histories, wage info, employer messages
Recommendation or guarantee randomization	trust, steering, platform credibility
Wage-floor or pay-rule experiments	price incidence, participation, bids, fill rates
Policy changes or redesigns	transparency, fees, governance, deactivation
Survey, conjoint, lab-in-field	fairness, flexibility, algorithm aversion, worker preferences

- Primary anchor: information and visibility for inexperienced workers in an online labor market.
- Challenge anchor: platform steering and guarantees; wage-rule extensions through transparency, compensation history, or minimum pay.
- Reproduce: compare treated and control workers in a synthetic platform panel.
- Diagnose: separate exposure, beliefs, wage changes, future reputation, risk, and welfare incidence.
- Transfer: redesign the empirical architecture for ranking, guarantees, wage floors, dynamic pricing, or governance.
- Deliverable: rule, labor margin, counterfactual, identification strategy, welfare object, and main threat.

Takeaways

- Platforms choose labor-market rules: visibility, assignment, prices, commissions, monitoring, and governance.
- The same rule can reduce frictions and shift surplus or risk.
- Applied work should identify the precise rule and the labor margin before making welfare claims.
- The frontier is empirical design: experiments, policy changes, platform data, and worker-preference evidence.